

MongoDB Lab – Day 6 (Advanced Queries)

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Roll No – CSE 326

Schema { emp_id, name, department, designation, city, salary, experience }

Initial Data:

Emp Id	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Kolkata	42000	1

Inserting Documents:

```
test> use employee_cse_326
switched to db employee_cse_326
employee_cse_326> db.employee_cse_326.insertMany([
... {emp_id:101,name:"Amit",department:"IT",designation:"Developer",city:"Kolkata",salary:60000,experience:3},
... {emp_id:102,name:"Neha",department:"HR",designation:"HR Executive",city:"Delhi",salary:45000,experience:2},
... {emp_id:103,name:"Ravi",department:"IT",designation:"Tester",city:"Mumbai",salary:50000,experience:4},
... {emp_id:104,name:"Pooja",department:"Finance",designation:"Accountant",city:"Chennai",salary:55000,experience:5},
... {emp_id:105,name:"Karan",department:"IT",designation:"Developer",city:"Kolkata",salary:65000,experience:6},
... {emp_id:106,name:"Sneha",department:"HR",designation:"Manager",city:"Delhi",salary:70000,experience:7},
... {emp_id:107,name:"Arjun",department:"Finance",designation:"Analyst",city:"Mumbai",salary:48000,experience:2},
... {emp_id:108,name:"Meera",department:"IT",designation:"Developer",city:"Bangalore",salary:75000,experience:8},
... {emp_id:109,name:"Rohit",department:"IT",designation:"Team Lead",city:"Pune",salary:90000,experience:10},
... {emp_id:110,name:"Anjali",department:"HR",designation:"Recruiter",city:"Kolkata",salary:42000,experience:1}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69abbb510c54ef8758628ca0'),
    '1': ObjectId('69abbb510c54ef8758628ca1'),
    '2': ObjectId('69abbb510c54ef8758628ca2'),
    '3': ObjectId('69abbb510c54ef8758628ca3'),
    '4': ObjectId('69abbb510c54ef8758628ca4'),
    '5': ObjectId('69abbb510c54ef8758628ca5'),
    '6': ObjectId('69abbb510c54ef8758628ca6'),
    '7': ObjectId('69abbb510c54ef8758628ca7'),
    '8': ObjectId('69abbb510c54ef8758628ca8'),
    '9': ObjectId('69abbb510c54ef8758628ca9')
  }
}
```

Lab Questions:

Q1. Display employee name and salary only.

Query: db.employee_cse_326.find({}, {name:1, salary:1, _id:0})

Output:

```
[
  { name: 'Amit', salary: 60000 },
  { name: 'Neha', salary: 45000 },
  { name: 'Ravi', salary: 50000 },
  { name: 'Pooja', salary: 55000 },
  { name: 'Karan', salary: 65000 },
  { name: 'Sneha', salary: 70000 },
  { name: 'Arjun', salary: 48000 },
  { name: 'Weera', salary: 75000 },
  { name: 'Rohit', salary: 90000 },
  { name: 'Anjali', salary: 42000 }
]
```

Q2. Find employees working in HR department.

Query: db.employee_cse_326.find({department:"HR"})

Output:

```
[
  {
    _id: ObjectId('69abbb510c54ef8758628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69abbb510c54ef8758628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69abbb510c54ef8758628ca9'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q3. Find employees whose salary is between 50,000 and 80,000.

Query: db.employee_cse_326.find({salary:{\$gte:50000,\$lte:80000}})

Output:

```
{
  {
    _id: ObjectId('69abb519c54ef8758628ca9'),
    emp_id: 181,
    name: 'Anil',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69abb519c54ef8758628ca2'),
    emp_id: 183,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69abb519c54ef8758628ca3'),
    emp_id: 184,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69abb519c54ef8758628ca4'),
    emp_id: 185,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  }
}
```

```
{
  {
    _id: ObjectId('69abb519c54ef8758628ca5'),
    emp_id: 186,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69abb519c54ef8758628ca7'),
    emp_id: 188,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
}
```

Q4. Display employees from Kolkata city.

Query: db.employee_cse_326.find({city:"Kolkata"})

Output:

```

{
  _id: ObjectId('69abb519c54ef8758628ca9'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('69abb519c54ef8758628ca4'),
  emp_id: 185,
  name: 'Naran',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('69abb519c54ef8758628ca9'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Malkata',
  salary: 42000,
  experience: 1
}
}

```

Q5. Sort employees by experience in ascending order.

Query: db.employee_cse_326.find().sort({experience:1})

Output:

```

{
  _id: ObjectId('69abb519c54ef8758628ca9'),
  emp_id: 110,
  name: 'Anjali',
  department: 'HR',
  designation: 'Recruiter',
  city: 'Malkata',
  salary: 42000,
  experience: 1
},
{
  _id: ObjectId('69abb519c54ef8758628ca1'),
  emp_id: 182,
  name: 'Nara',
  department: 'HR',
  designation: 'HR Executive',
  city: 'Malkata',
  salary: 45000,
  experience: 2
},
{
  _id: ObjectId('69abb519c54ef8758628ca5'),
  emp_id: 187,
  name: 'Ajay',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumail',
  salary: 48000,
  experience: 2
},
{
  _id: ObjectId('69abb519c54ef8758628ca9'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 60000,
  experience: 3
},
{
  _id: ObjectId('69abb519c54ef8758628ca2'),
  emp_id: 183,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumail',
  salary: 50000,
  experience: 4
},
}

```

```

{
  _id: ObjectId('69abb519c54ef8758628ca3'),
  emp_id: 184,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 5
},
{
  _id: ObjectId('69abb519c54ef8758628ca4'),
  emp_id: 185,
  name: 'Naran',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 65000,
  experience: 6
},
{
  _id: ObjectId('69abb519c54ef8758628ca5'),
  emp_id: 186,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Malkata',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('69abb519c54ef8758628ca7'),
  emp_id: 188,
  name: 'Meera',
  department: 'IT',
  designation: 'Developer',
  city: 'Bangalore',
  salary: 75000,
  experience: 8
},
{
  _id: ObjectId('69abb519c54ef8758628ca8'),
  emp_id: 189,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Mumail',
  salary: 80000,
  experience: 10
}
}

```

Q6. Display top 5 most experienced employees.

Query: `db.employee_cse_326.find().sort({experience:-1}).limit(5)`

Output:

```
{
  "_id": ObjectId("69abbb518c54ef8758628ca8"),
  "emp_id": 189,
  "name": "Rishi",
  "department": "IT",
  "designation": "Team Lead",
  "city": "Bangalore",
  "salary": 80000,
  "experience": 10
},
{
  "_id": ObjectId("69abbb518c54ef8758628ca7"),
  "emp_id": 188,
  "name": "Rishi",
  "department": "IT",
  "designation": "Developer",
  "city": "Bangalore",
  "salary": 75000,
  "experience": 8
},
{
  "_id": ObjectId("69abbb518c54ef8758628ca5"),
  "emp_id": 186,
  "name": "Sheha",
  "department": "HR",
  "designation": "Manager",
  "city": "Delhi",
  "salary": 70000,
  "experience": 9
},
{
  "_id": ObjectId("69abbb518c54ef8758628ca4"),
  "emp_id": 185,
  "name": "Rishi",
  "department": "IT",
  "designation": "Developer",
  "city": "Mumbai",
  "salary": 60000,
  "experience": 6
},
{
  "_id": ObjectId("69abbb518c54ef8758628ca3"),
  "emp_id": 184,
  "name": "Rishi",
  "department": "Finance",
  "designation": "Accountant",
  "city": "Chennai",
  "salary": 55000,
  "experience": 5
}
```

Q7. Update salary of employee with emp_id 102 to 48,000.

Query: `db.employee_cse_326.updateOne({emp_id:102},{set:{salary:48000}})`

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Increase salary of all IT employees by 10,000.

Query: `db.employee_cse_326.updateMany({department:"IT"},{$inc:{salary:10000}})`

Output:

```
employee_cse_326> db.employee_cse_326.updateMany({department:"IT",experience:{$gt:5}},{$set:{designation:"Senior Developer"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Q9. Update designation to “Senior Developer” for IT employees having experience more than 5 years.

Query:

```
db.employee_cse_326.updateMany({department:"IT",experience:{$gt:5}},{$set:{designation:"Senior Developer"}})
```

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q10. Delete employees whose experience is less than 2 years.

Query: db.employee_cse_326.deleteMany({experience:{\$lt:2}})

Output:

```
employee_cse_326> db.employee_cse_326.deleteMany({experience:{$lt:2}})
{ acknowledged: true, deletedCount: 1 }
```

Q11. Display employees who are not from Delhi.

Query: db.employee_cse_326.find({city:{\$ne:"Delhi"}})

Output:

```
{
  "_id": ObjectId("69abb518c54ef8758628ca9"),
  "emp_id": 181,
  "name": "Amit",
  "department": "IT",
  "designation": "Developer",
  "city": "Molkata",
  "salary": 78000,
  "experience": 3
},
{
  "_id": ObjectId("69abb518c54ef8758628ca2"),
  "emp_id": 182,
  "name": "Ravi",
  "department": "IT",
  "designation": "Tester",
  "city": "Mumbai",
  "salary": 60000,
  "experience": 4
},
{
  "_id": ObjectId("69abb518c54ef8758628ca3"),
  "emp_id": 184,
  "name": "Pooja",
  "department": "Finance",
  "designation": "Accountant",
  "city": "Chennai",
  "salary": 55000,
  "experience": 5
},
{
  "_id": ObjectId("69abb518c54ef8758628ca4"),
  "emp_id": 185,
  "name": "Naran",
  "department": "IT",
  "designation": "Senior Developer",
  "city": "Molkata",
  "salary": 75000,
  "experience": 6
}
},
{
  "_id": ObjectId("69abb518c54ef8758628ca6"),
  "emp_id": 187,
  "name": "Anand",
  "department": "Finance",
  "designation": "Analyst",
  "city": "Mumbai",
  "salary": 48000,
  "experience": 2
},
{
  "_id": ObjectId("69abb518c54ef8758628ca7"),
  "emp_id": 188,
  "name": "Meera",
  "department": "IT",
  "designation": "Senior Developer",
  "city": "Bangalore",
  "salary": 85000,
  "experience": 8
},
{
  "_id": ObjectId("69abb518c54ef8758628ca8"),
  "emp_id": 189,
  "name": "Rohit",
  "department": "IT",
  "designation": "Senior Developer",
  "city": "Pune",
  "salary": 100000,
  "experience": 10
}
}
```

Q12. Find employees whose designation is Developer and salary greater than 60,000.

Query: db.employee_cse_326.find({designation:"Developer",salary:{\$gt:60000}})

Output:

```
{
  "_id": ObjectId("69abb518c54ef8758628ca9"),
  "emp_id": 181,
  "name": "Amit",
  "department": "IT",
  "designation": "Developer",
  "city": "Molkata",
  "salary": 78000,
  "experience": 3
}
}
```

Q13. Display employees working in IT or Finance department.

Query: db.employee_cse_326.find({\$or:[{department:"IT"},{department:"Finance"}]}))

Output:

```

{
  _id: ObjectId('69abb518c54ef8758628ca0'),
  emp_id: 181,
  name: 'Amit',
  department: 'IT',
  designation: 'Developer',
  city: 'Malkata',
  salary: 70000,
  experience: 3
},
{
  _id: ObjectId('69abb518c54ef8758628ca2'),
  emp_id: 183,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 60000,
  experience: 4
},
{
  _id: ObjectId('69abb518c54ef8758628ca3'),
  emp_id: 184,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 5
},
{
  _id: ObjectId('69abb518c54ef8758628ca4'),
  emp_id: 185,
  name: 'Murali',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Malkata',
  salary: 75000,
  experience: 6
},
}

```

```

{
  _id: ObjectId('69abb518c54ef8758628ca6'),
  emp_id: 187,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 40000,
  experience: 2
},
{
  _id: ObjectId('69abb518c54ef8758628ca7'),
  emp_id: 188,
  name: 'Meera',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Bangalore',
  salary: 85000,
  experience: 8
},
{
  _id: ObjectId('69abb518c54ef8758628ca8'),
  emp_id: 189,
  name: 'Rohit',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Pune',
  salary: 100000,
  experience: 10
},
}

```

Q14. Display employees sorted by department and then by salary.

Query: `db.employee_cse_326.find().sort({department:1,salary:1})`

Output:

```

{
  _id: ObjectId('69abb518c54ef8758628ca6'),
  emp_id: 187,
  name: 'Arjun',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 40000,
  experience: 2
},
{
  _id: ObjectId('69abb518c54ef8758628ca3'),
  emp_id: 184,
  name: 'Pooja',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Chennai',
  salary: 55000,
  experience: 5
},
{
  _id: ObjectId('69abb518c54ef8758628ca1'),
  emp_id: 182,
  name: 'Neha',
  department: 'HR',
  designation: 'HR Executive',
  city: 'Delhi',
  salary: 60000,
  experience: 2
},
{
  _id: ObjectId('69abb518c54ef8758628ca5'),
  emp_id: 186,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
}

```



```

{
  _id: ObjectId('69abbb510c54ef8758628ca2'),
  emp_id: 183,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 60000,
  experience: 4
},
{
  _id: ObjectId('69abbb510c54ef8758628ca3'),
  emp_id: 181,
  name: 'Ravi',
  department: 'IT',
  designation: 'Developer',
  city: 'Mumbai',
  salary: 70000,
  experience: 3
},
{
  _id: ObjectId('69abbb510c54ef8758628ca4'),
  emp_id: 182,
  name: 'Ravi',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Mumbai',
  salary: 75000,
  experience: 6
},
{
  _id: ObjectId('69abbb510c54ef8758628ca7'),
  emp_id: 180,
  name: 'Ravi',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Bangalore',
  salary: 85000,
  experience: 8
},
{
  _id: ObjectId('69abbb510c54ef8758628ca8'),
  emp_id: 189,
  name: 'Ravi',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Mumbai',
  salary: 100000,
  experience: 10
}
}

```

Q15. Count total number of employees in the company.

Query: `db.employee_cse_326.countDocuments()`

Output:

```

employee_cse_326> db.employee_cse_326.countDocuments()
9

```

Q16. Display employees whose salary is not equal to 50,000.

Query: `db.employee_cse_326.find({salary:{$ne:50000}})`

Output:

```

{
  _id: ObjectId('69abbb510c54ef8758628ca3'),
  emp_id: 181,
  name: 'Ravi',
  department: 'IT',
  designation: 'Developer',
  city: 'Mumbai',
  salary: 70000,
  experience: 3
},
{
  _id: ObjectId('69abbb510c54ef8758628ca1'),
  emp_id: 182,
  name: 'Ravi',
  department: 'HR',
  designation: 'HR Executive',
  city: 'Mumbai',
  salary: 40000,
  experience: 2
},
{
  _id: ObjectId('69abbb510c54ef8758628ca2'),
  emp_id: 183,
  name: 'Ravi',
  department: 'IT',
  designation: 'Tester',
  city: 'Mumbai',
  salary: 60000,
  experience: 4
},
{
  _id: ObjectId('69abbb510c54ef8758628ca3'),
  emp_id: 184,
  name: 'Ravi',
  department: 'Finance',
  designation: 'Accountant',
  city: 'Mumbai',
  salary: 55000,
  experience: 5
}
}

```

```

{
  _id: ObjectId('69abbb510c54ef8758628ca4'),
  emp_id: 100,
  name: 'Narrah',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Kolkata',
  salary: 75000,
  experience: 6
},
{
  _id: ObjectId('69abbb510c54ef8758628ca5'),
  emp_id: 106,
  name: 'Sneha',
  department: 'HR',
  designation: 'Manager',
  city: 'Delhi',
  salary: 70000,
  experience: 7
},
{
  _id: ObjectId('69abbb510c54ef8758628ca6'),
  emp_id: 107,
  name: 'Ajay',
  department: 'Finance',
  designation: 'Analyst',
  city: 'Mumbai',
  salary: 48000,
  experience: 2
},
{
  _id: ObjectId('69abbb510c54ef8758628ca7'),
  emp_id: 108,
  name: 'Rishi',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Bangalore',
  salary: 85000,
  experience: 8
},
{
  _id: ObjectId('69abbb510c54ef8758628ca8'),
  emp_id: 109,
  name: 'Rishi',
  department: 'IT',
  designation: 'Senior Developer',
  city: 'Pune',
  salary: 100000,
  experience: 10
}
]

```

▶ **_id: ObjectId('69abbb510c54ef8758628ca0')**
 emp_id : 101
 name : "Amit"
 department : "IT"
 designation : "Developer"
 city : "Kolkata"
 salary : 70000
 experience : 3

_id: ObjectId('69abbb510c54ef8758628ca1')
 emp_id : 102
 name : "Neha"
 department : "HR"
 designation : "HR Executive"
 city : "Delhi"
 salary : 48000
 experience : 2

_id: ObjectId('69abbb510c54ef8758628ca2')
 emp_id : 103
 name : "Ravi"
 department : "IT"
 designation : "Tester"
 city : "Mumbai"
 salary : 60000
 experience : 4

_id: ObjectId('69abbb510c54ef8758628ca3')
 emp_id : 104
 name : "Pooja"
 department : "Finance"
 designation : "Accountant"
 city : "Chennai"
 salary : 55000
 experience : 5

_id: ObjectId('69abbb510c54ef8758628ca4')
 emp_id : 105
 name : "Karan"
 department : "IT"
 designation : "Senior Developer"
 city : "Kolkata"
 salary : 75000
 experience : 6

_id: ObjectId('69abbb510c54ef8758628ca5')
 emp_id : 106
 name : "Sneha"
 department : "HR"
 designation : "Manager"
 city : "Delhi"
 salary : 70000
 experience : 7

```
_id: ObjectId('69abbb510c54ef8758628ca6')
emp_id: 107
name: "Arjun"
department: "Finance"
designation: "Analyst"
city: "Mumbai"
salary: 48000
experience: 2
```

```
_id: ObjectId('69abbb510c54ef8758628ca7')
emp_id: 108
name: "Meera"
department: "IT"
designation: "Senior Developer"
city: "Bangalore"
salary: 85000
experience: 8
```

```
_id: ObjectId('69abbb510c54ef8758628ca8')
emp_id: 109
name: "Rohit"
department: "IT"
designation: "Senior Developer"
city: "Pune"
salary: 100000
experience: 10
```